

DARK PATTERNS BUSTER HACKATHON-2023

The Dark Forces: A Web Extension Solution

SUBMITTED BY

AYUSH YADAV (AD-1228)

RIYANSH SHARMA (AD-1607)

SHRISHTI RAWAT (AD-1289)

SHUBH SANKHYADHAR (AD-1290)

B.Sc. (HONS) COMPUTER SCIENCE

ACHARYA NARENDRA DEV COLLEGE
UNIVERSITY OF DELHI

Contents

1	INTRODUCTION	3
1.1	Overview	3
1.2	Objectives	3
2	Getting Started	5
2.1	Supported Browsers	5
2.2	Installation	5
2.3	Usage	6
3	TOOLS AND TECHNOLOGY USED	7
3.1	Technologies used in Development	7
3.1.1	For Frontend Development	7
3.1.2	For Backend Development	7
3.1.3	For Machine Learning	8
3.1.4	For server-side operations	8
3.1.5	The Dataset	8
4	Workflow Overview	9
4.1	Web exploration	9
4.2	Categorization	9
4.3	Analysing the website	10
4.4	ML Analysis and Predictions	11
4.5	Highlighting and Frequency report generation	12
4.6	Extension's Interface	13
5	Features	14
5.1	Dark Patterns detected	14
5.2	Non-intrusive interaction while web page analysis	15
5.3	Intelligent color highlighting and Frequency report generation	16
5.4	Voice Alerts	17
5.5	Toggle Buttons for stopping extension on a sepcific website	17

5.6	Optimized for chromium browsers	18
5.7	Respecting user's privacy and website's terms of service	18
5.8	Extension detects dark patterns with high accuracy and low latency	18
6	CONCLUSION AND FUTURE SCOPE	19
6.1	Conclusion	19
6.2	Future Scope	20
6.2.1	Increased accuracy with good computational resources.	20
6.2.2	Database integration to tackle variable nature of dark patterns	20
6.2.3	Making suitable for non-chromium based browsers . . .	20
6.2.4	Scaling up for detecting more dark patterns	21

Chapter 1

INTRODUCTION

In the dynamic realm of online commerce, user experiences often encounter challenges posed by deceptive design practices, commonly known as dark patterns. As we navigate the digital landscape, the prevalence of these manipulative strategies calls for innovative solutions that empower users with heightened awareness and control during their online interactions. It is within this context that we present The Dark Forces, an advanced web extension meticulously crafted to tackle the intricacies of dark patterns on e-commerce platforms.

1.1 Overview

The Dark Forces is more than a tool; it's a strategic response to the complexities of online shopping. Utilizing technologies like machine learning models and semantic analysis, our extension acts as a vigilant guardian, detecting, categorizing, and highlighting of dark patterns in real-time. This proactive approach aims to foster a transparent, secure, and user-centric online environment. At its core, The Dark Forces aims to restore user agency and confidence in online shopping. By diligently identifying and flagging instances of dark patterns, the extension provides users with a shield against manipulative design, promoting an online ecosystem characterized by trust, clarity, and informed decision-making.

1.2 Objectives

- Develop a sophisticated algorithm to discern and highlight instances of dark patterns with precision and accuracy.

- Implement a responsive system that delivers real-time notifications and visually intuitive cues to users, fostering heightened awareness during their online interactions..
- Prioritize the security and privacy of user data throughout the entire operational lifecycle, from data extraction to secure communication with backend servers.
- Integrate supplementary features such as voice alerts and toggle buttons, enhancing user interaction and contributing to an improved overall online experience..

Chapter 2

Getting Started

Dark Forces, a sophisticated browser extension designed to identify and mitigate dark patterns on e-commerce websites. Dark Forces empowers users by providing insights into deceptive design practices, ensuring a more transparent and ethical online shopping experience.

2.1 Supported Browsers

To experience the full benefits of the extension Dark Forces, it's essential to use a Chromium-based browser. Currently, our extension seamlessly integrates with popular browsers like Google Chrome, Opera Mini, Brave, and Microsoft Edge. Chromium-based browsers provide a robust platform for the extension to deliver effective dark pattern detection. Unfortunately, non-Chromium browsers like Safari or Mozilla Firefox are not currently supported.

2.2 Installation

- i Browser Compatibility: Ensure your browser is a Chromium-based application, such as Google Chrome, Opera Mini, Brave, or Microsoft Edge. Currently, this extension does not support non-Chromium browsers like Safari or Mozilla Firefox.
- ii Enabling the extension: To enable the Dark Pattern Detection Web Extension, follow these steps:
 - Download the extension package .
 - Unzip the package.

- Open your browser and go to the extensions page.
- Enable "Developer Mode."
- Click "Load Unpacked" and select the unzipped extension folder.

2.3 Usage

- Comprehensive Solution: This extension enhances users' online browsing experience by identifying and highlighting potential dark patterns on e-commerce websites.
- Intelligent operation: Operates intelligently in the background as users navigate e-commerce platforms. Automatically analyzes web pages for design elements indicative of dark patterns.
- User-Friendly Alerts: Provides real-time alerts and visual cues to draw attention to identified dark patterns.
- User Contribution: Users can contribute to the extension's improvement by providing feedback on detected patterns.
- Customization Options: Empowers users with customization options through extension settings.

Chapter 3

TOOLS AND TECHNOLOGY USED

In crafting this web extension, a careful selection of cutting-edge tools and technologies was pivotal to ensuring its efficiency, accuracy, and user-friendliness. This section provides an insightful overview of the tools and technologies harnessed during the development process. Each element was chosen to enhance the extension's capabilities, making it a robust solution for identifying and mitigating dark patterns on e-commerce websites. Let's delve into the technological foundations that power this extension and contribute to its seamless functionality.

3.1 Technologies used in Development

3.1.1 For Frontend Development

Purpose: Crafting an intuitive and visually appealing interface for this extension.

Details: The use of HTML, CSS, and JavaScript ensured the development of a seamless front end experience, enabling users to interact effortlessly with the extension.

3.1.2 For Backend Development

Purpose: Handling the extraction of content from websites and orchestrating communication with the machine learning models.

Details: Leveraging JavaScript in the back end facilitated the extraction of

website content, initiating predictions from the machine learning models, and coordinating the highlighting of dark patterns in the user interface.

3.1.3 For Machine Learning

The machine learning backbone of this extension is built upon the robust capabilities of Python. The advanced approach of Ensembled modelling is used. Leveraging the versatility and efficiency of the Python language, we have meticulously trained three distinct LLMs—Bert, Roberta and XLNet—on a carefully curated dataset. Further an enhanced dataset is created based on the predictions of these models. Roberta is further trained on that enhanced dataset which serve as the predictive engine, discerning potential dark patterns lurking within online interfaces makes final predictions.

3.1.4 For server-side operations

Powering the back end infrastructure of this extension is the swift and efficient Python Flask. Tasked with processing data and triggering model predictions, Flask proves to be a strategic choice for its notable speed and responsiveness. The adoption of Flask enhances the extension’s overall performance, ensuring seamless data handling and rapid execution of essential functionalities. This strategic implementation contributes to the extension’s agility in processing information and delivering timely predictions, further elevating the user experience.

3.1.5 The Dataset

The foundation of our machine learning models lies in the dataset generously provided by IIT BHU workshop [1], a cornerstone established during the Dark Patterns Buster Hackathon. Leveraging this meticulously curated dataset, our models—Bert, Roberta and XLNet—have undergone rigorous training, using which new enhanced dataset is generated on which a new meta model is trained to equip it with the proficiency to detect dark patterns in online environments. The dataset’s quality, sourced from the expertise of IIT BHU, has played a pivotal role in shaping the accuracy and reliability of our models. This collaborative effort reflects the commitment to advancing ethical online practices, championed by both academia and industry collaboration.

Chapter 4

Workflow Overview

Navigating the intricate landscape of dark pattern detection involves a meticulously orchestrated sequence of actions. From the initiation of the extension to the final presentation of detected patterns, each step in the workflow is designed with precision and efficiency. Explore the detailed processes that define the workflow of The Dark Forces as it seamlessly detects, analyzes, and empowers users in the dynamic realm of online commerce.

4.1 Web exploration

Upon user activation of The Dark Forces web extension, a seamless communication flow is initiated. The content script sends a message to the background, which, acting as a service worker, records the active tab. This meticulous orchestration ensures that the extension is aware of the user's current browsing context.

4.2 Categorization

Our extension employs an effective mechanism to categorize web pages into either e-commerce or non-e-commerce domains as users navigate the internet. Initially, it relies on a predefined list of well-known e-commerce websites to facilitate categorization. However, to enhance scalability, the extension is equipped with the capability to extract the Document Object Model (DOM) structure of each visited web page. Utilizing sophisticated algorithms such as Naive Bayes, the extension can intelligently categorize websites based on their inherent characteristics.

When the extension identifies a web page as belonging to the e-commerce category, it initiates further in-depth analysis. Conversely, if a web page is determined to be non-e-commerce, the extension refrains from extracting its DOM structure, optimizing the efficiency of the categorization process. This approach ensures that the extension focuses its resources on relevant e-commerce websites, streamlining the analysis and prioritizing meaningful insights for the user.

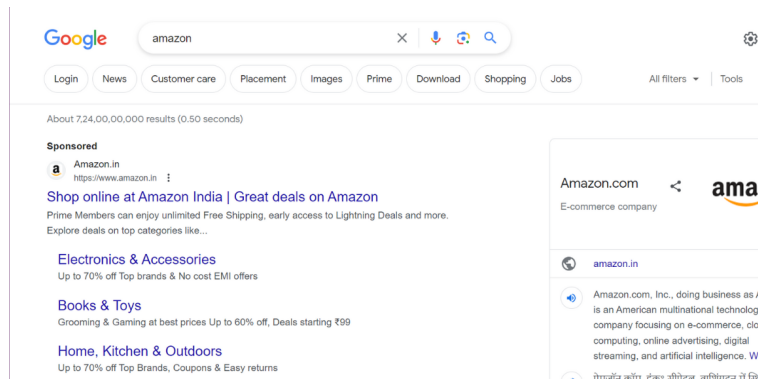


Figure 4.1: Not Working on non- e-commerce website

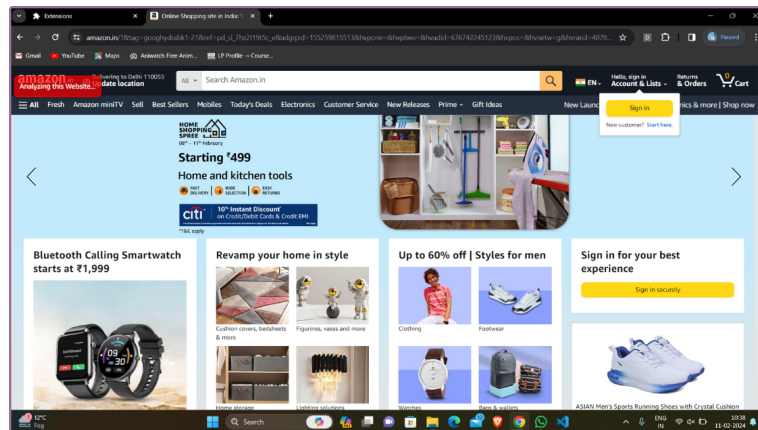


Figure 4.2: Working on e-commerce website

4.3 Analysing the website

Once extension finds any e-commerce website, it engages in meticulous data extraction. Parsing the entire body tag of the webpage, the script traverses

each HTML tag, selectively capturing textual content. Non-textual tags, such as script, style, hr, and br, are intelligently excluded, as they bear no relevance to the identification of dark patterns. During analysis of website, user's interaction with the website does not get interrupted. The extracted text is then carefully manipulated and organized into an array using some data pre-processing methods to make the text ready for predictive analysis.

Segmentation and Filtering (Segment.js): Defines functions for dividing the Document Object Model (DOM) into meaningful segments. Filters out specific types of elements (e.g., ignored elements, pixels, hidden elements) to streamline analysis.

Common Utilities (Common.js): Houses utility functions for various purposes, such as calculating element areas, checking visibility, and determining inter-actability.

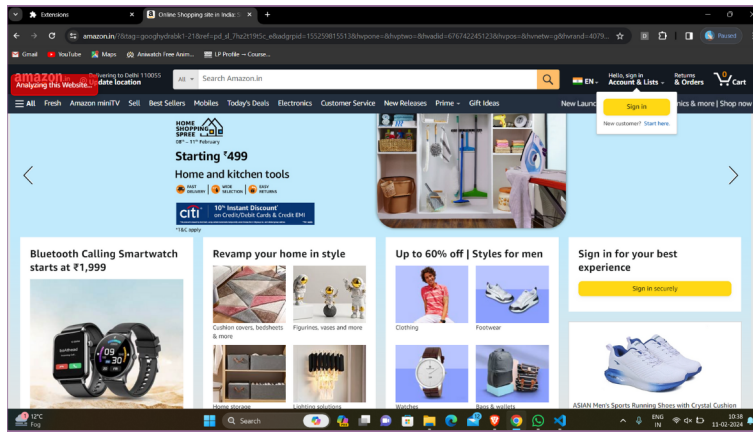


Figure 4.3: Analysing e-commerce website

4.4 ML Analysis and Predictions

The array of text content undergoes a transformative journey as it is dispatched to a backend Flask server. This server activates the ensembled Roberta Model which predicts the presence of dark patterns. Semantic and sentiment analysis is applied to refine the predictions.

4.5 Highlighting and Frequency report generation

The insights gleaned from the machine learning models, presented in the form of labels and predictions, embark on their return journey. The backend server communicates this invaluable data back to the content script, establishing a bridge between analysis and action. The content script, armed with these predictions, cross-references them with the elements array, effectively pinpointing and highlighting dark patterns on the webpage. Different colors are used for different dark patterns. This intricate communication ensures that the user is promptly made aware of potentially deceptive design elements. Detection is not the final act; it is the catalyst for user enlightenment. Upon successful identification of dark patterns, a frequency report of dark patterns found on website with a pie chart is generated. To augment this informative experience, voice alerts accompany the message, providing an additional layer of awareness.

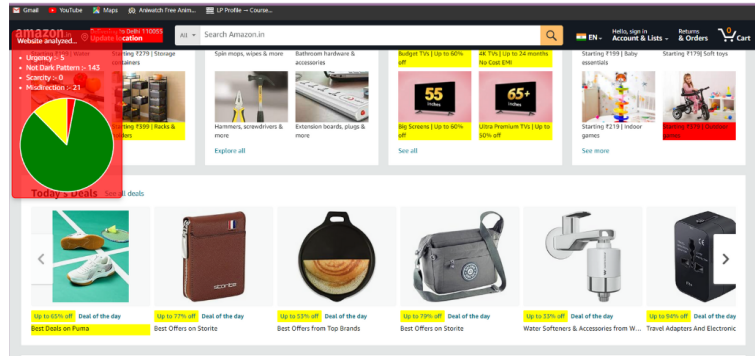


Figure 4.4: Highlighting and frequency report generation

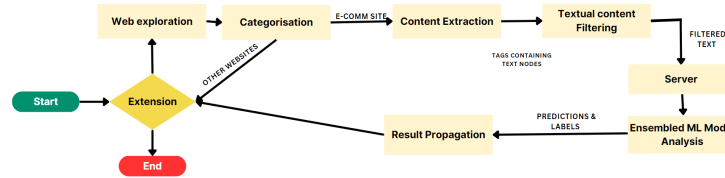


Figure 4.5: Work Flow Chart

4.6 Extension's Interface

Furthermore, two toggle buttons grace the extension's interface which can be seen by clicking on extension icon. This interface grant users the ability to stop the extension for a particular website in future browsing and second is to disable the extension on all websites. This interface also provides the user with the information of dark patterns it can detect.

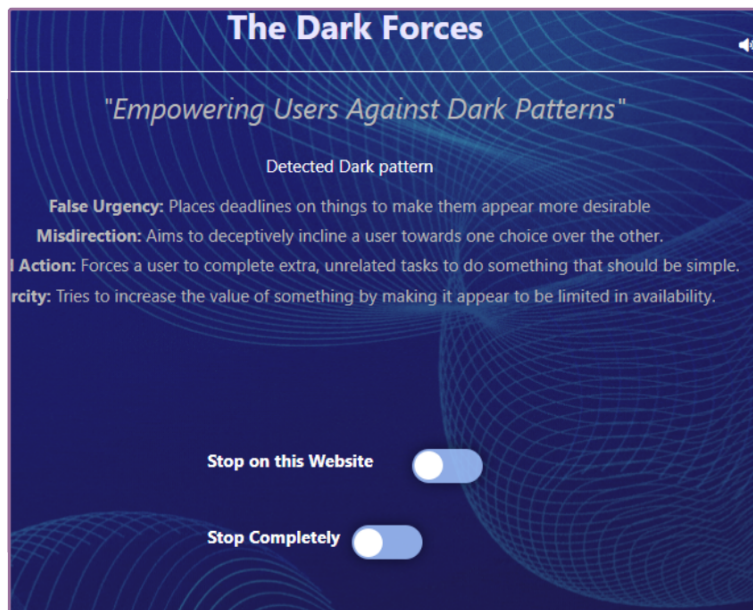


Figure 4.6: Extension's Interface

Chapter 5

Features

The user interaction component of The Dark Forces is a thoughtfully crafted experience aimed at seamlessly engaging and empowering users throughout their online journey.

5.1 Dark Patterns detected

Currently, our extension is able to detect three dark patterns namely:

Misdirection: Misdirection involves guiding users' attention away from the actual consequences or actions, often leading them to make choices they might not have otherwise made. It can involve hiding information or presenting it in a way that diverts attention.

Scarcity: Scarcity dark pattern creates a sense of limited availability, pushing users to take immediate actions by making them believe that a product or offer is scarce. This can lead to impulsive decisions without thorough consideration.

False Urgency: False urgency involves creating a sense of urgency through misleading cues or time constraints, pressuring users to make quick decisions. This tactic aims to exploit the fear of missing out (FOMO) and encourages hasty actions that users might later regret.

Extension can be extended for detecting more dark patterns based on the similar approach of Ensembled modelling used for above dark patterns detection.

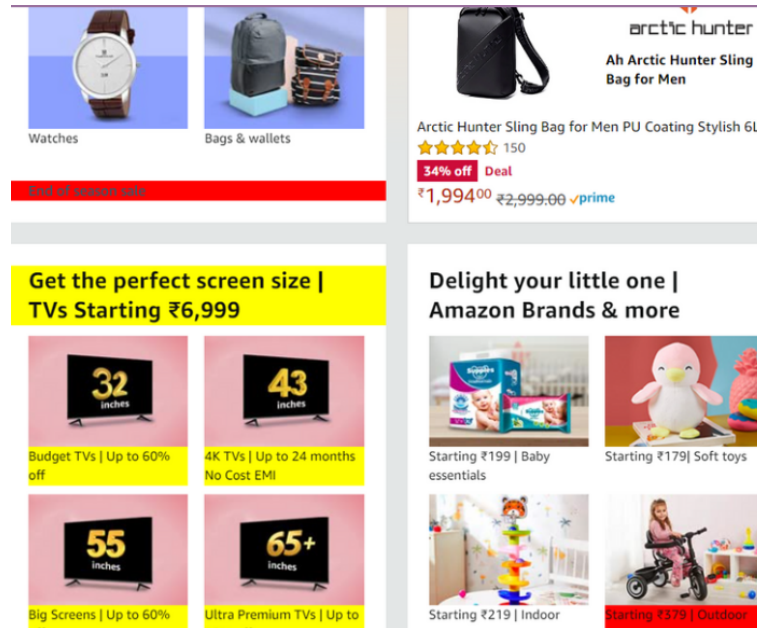


Figure 5.1: Red Color Showing Urgency and Yellow Color Showing Misdirection

5.2 Non-intrusive interaction while web page analysis

User interaction initiates with the extension popup, triggered effortlessly upon the detection of dark patterns. This unintrusive notification serves as a gateway, inviting users to delve into a detailed exploration of identified issues, ensuring a timely and user-centric response.

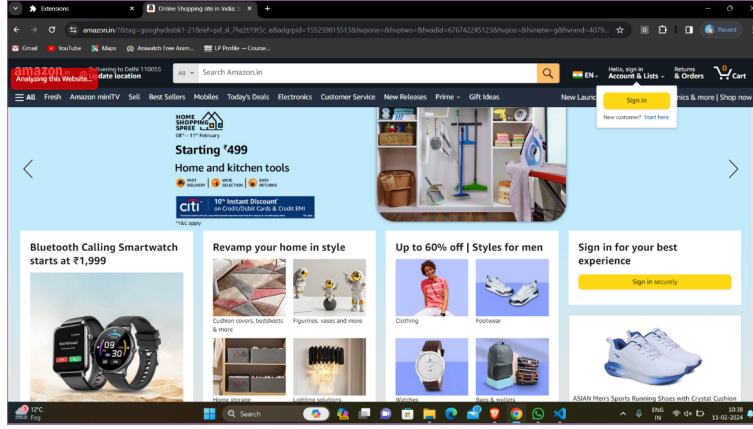


Figure 5.2: Extension Triggered

5.3 Intelligent color highlighting and Frequency report generation

Within the extension popup, a comprehensive narrative unfolds. Users are presented with an informative display showcasing the specifics of detected dark patterns with a pie chart. This transparency not only educates users on potential pitfalls in the online content but also instill a sense of awareness and informed decision-making.

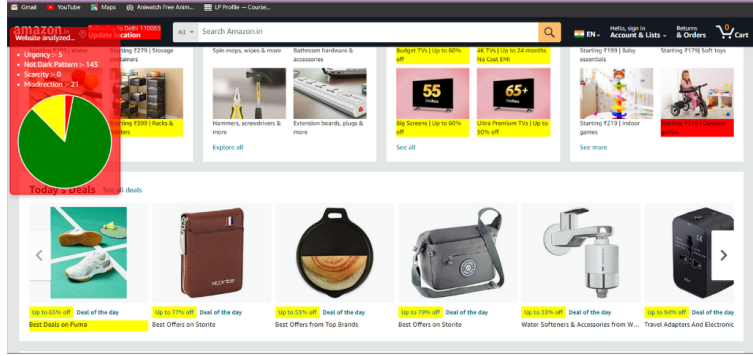


Figure 5.3: Overall highlighting and frequency report generation

5.4 Voice Alerts

To elevate the user experience, The Dark Forces introduces voice alerts. These auditory companions, seamlessly integrated with the extension popup, provide an additional layer of engagement. Voice alerts serve as a proactive means of drawing immediate attention to the presence of dark patterns, enhancing user awareness in a non-intrusive manner.

5.5 Toggle Buttons for stopping extension on a sepcific website

In the spirit of user empowerment, the extension popup features two intuitive toggle buttons. The first allows users to stop extension for a particular website. The second toggle button empowers users to stop extension for all websites. Further, voice alerts can also be stopped by clicking the sound icon in top right corner.

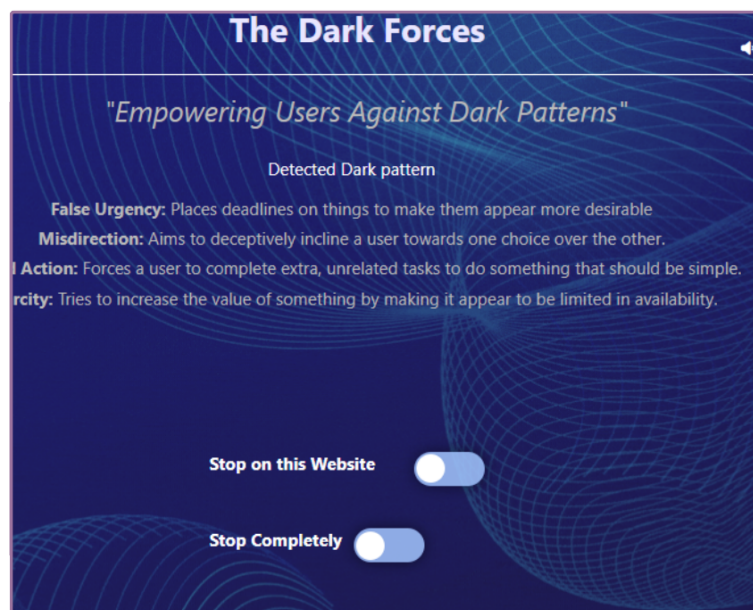


Figure 5.4: Toggle Buttons

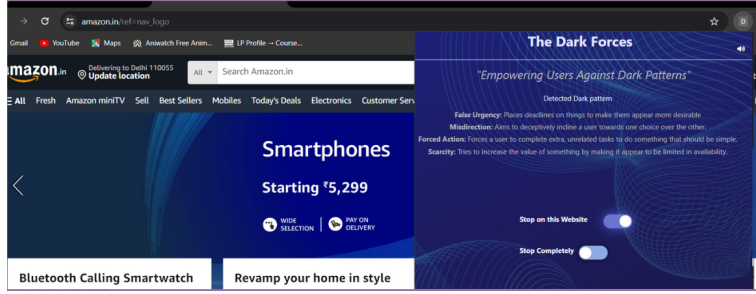


Figure 5.5: Stop extension for a particular website functionality

5.6 Optimized for chromium browsers

To experience the full benefits of the extension Dark Forces, it's essential to use a Chromium-based browser. Currently, our extension seamlessly integrates with popular browsers like Google Chrome, Opera Mini, Brave, and Microsoft Edge. Chromium-based browsers provide a robust platform for the extension to deliver effective dark pattern detection. Unfortunately, non-Chromium browsers like Safari or Mozilla Firefox are not currently supported. In unison, these elements of user interaction in The Dark Forces converge to create an immersive and responsive user experience, redefining the user's role in the ongoing battle against dark patterns.

5.7 Respecting user's privacy and website's terms of service

Our extension does not store any information of user thereby protecting the privacy of user. Further, our extension, does not manipulate the web page's content thereby respecting website's terms of service.

5.8 Extension detects dark patterns with high accuracy and low latency

Our extension has achieved an accuracy of 97.68 in detecting dark pattern with just a latency of 30 seconds making itself the most suitable solution which incorporates the accuracy of LLMs but does not get affected by their large size.

Chapter 6

CONCLUSION AND FUTURE SCOPE

This chapter explains the conclusion followed by the future scope of the done work.

6.1 Conclusion

The development of The Dark Forces web extension represents a dedicated effort to confront the challenges presented by dark patterns in online commerce.

From its inception, the extension has evolved into a strategic response to the complexities of online shopping. Leveraging cutting-edge technologies, including advanced machine learning models, an advanced approach of Ensembled modelling it serves as a vigilant guardian—detecting, categorizing, and mitigating the impact of dark patterns in real-time. The overarching goal is to foster a transparent, secure, and user-centric online environment, restoring user agency and confidence.

Our detailed exploration of the extension’s workflow highlighted its intricate processes, from communication flow to data extraction, machine learning model integration, and seamless communication between the server and content script. The emphasis on user interaction, with features like extension popups, voice alerts, and toggle buttons, underscores our commitment to providing a comprehensive and user-friendly solution.

6.2 Future Scope

As we look ahead, the future scope of The Dark Forces holds exciting possibilities for further innovation and impact. The following points delve into the potential areas of growth and improvement that will shape the next phase of our project which are not completed yet due to the time constraint we had.

6.2.1 Increased accuracy with good computational resources.

Expanding our extension’s capabilities involves enhancing machine learning models through extensive training with a sizable dataset and robust computational resources. Despite, having system server, it achieved a good accuracy and low latency which can be further improved if proper computational resources are provided.

6.2.2 Database integration to tackle variable nature of dark patterns

Currently, our solution is able to differentiate between a dark pattern and a genuine sentence flashing on website. But, considering the variable nature of dark patterns, this needs to be more robust. For that, we are planning to maintain a database, which will be inputted with number of dark patterns found with website’s name every time when user navigates an e-commerce website. A threshold of dark patterns will be set. Crossing that threshold will state that website is using a lot of dark patterns. This will help in a way, when any user visits a website, last 5 entries for that website in database will be checked whether maximum number of times, website crossed threshold of dark patterns or not. This will be considered together with predictions of dark patterns. If a website crossed threshold of dark patterns, maximum no. of times then the predicted dark pattern by the model will be considered. In this way, the approach of reinforcement learning can be used to increase robustness of the solution where solution will learn each time and improve its dataset each time user navigates the web page. This will help in detecting whether a sentence is a dark pattern or not, more accurately.

6.2.3 Making suitable for non-chromium based browsers

In our pursuit of continuous improvement, we envision extending the compatibility of our extension beyond Chromium-based browsers. This expansion will enable our cutting-edge dark pattern detection technology on non-

Chromium platforms, ensuring a broader reach and enhanced user experience across diverse web ecosystems.

6.2.4 Scaling up for detecting more dark patterns

This extension can be scaled up using the similar approach for detecting more dark patterns.

In this phase of future exploration, our focus remains on continuous improvement, adaptability, and a commitment to providing users with a seamless and secure online environment. As we venture into these exciting possibilities, we aim to redefine the standards of transparency and user-centricity in the evolving landscape of online commerce.

Bibliography

- [1] Feng J. Matsumoto T. Fukushima N. Kido F. Yamana H. (Year). Yada, Y. Dark patterns in e-commerce: a dataset and its baseline evaluations. Journal Name, Volume(Issue), 12 Nov 2022.